

1. The experiment and the concentration we want

A 550 km LEO satellite directly overhead illuminates one atmospheric column. We look for extra acetonitrile (CH_3CN).

What the receiver is looking for

Gas molecules remove different amounts of power at different frequencies. We estimate how much the known CH_3CN absorption pattern has increased between two acquisitions.

Gas enhancement means $\Delta c(z) = c_{\text{sample}}(z) - c_{\text{reference}}(z)$: extra CH_3CN above the reference atmosphere. It is simulated here, rather than measured pollution.

$$\Delta c(z) = q e^{-z/(1500 \text{ m})}$$

q is the surface equivalent concentration increase ($\mu\text{g}/\text{m}^3$). The fixed 1,500 m profile lets one scalar describe the column; one ray cannot recover concentration at every height.

Example input: $q = 50.5814 \mu\text{g}/\text{m}^3$ ($18.6079 \mu\text{g}/\text{m}^3$ at 1,500 m). It was chosen for 95% predicted detection with an assumed 0.001 dB calibration residual. The receiver estimates q without knowing this input.

Actual HITRAN data and ITU atmosphere models support the calculation. The gas profile, hardware and receiver observations are modeled.

The specified link

Setting	Value and meaning
Elevation	90° at the receiver, shortest geometric path
Band and tones	230–240 GHz across 1,024 tones
RF power	25 dBm total average power across the band
Apertures	0.10 m transmit / 1.0 m receive, efficiency 0.65
Receiver	7 dB noise figure, 5 dB implementation loss
Acquisition	10 s reference + 10 s sample

HITRAN supplies molecular line data, not the concentration q . This example chooses q using page 6's sensitivity calculation. A 3D image would need multiple rays and additional assumptions.

2. Molecules, height and optical depth

The forward model adds the absorption of each altitude layer. Its inputs are a concentration profile and spectral line parameters.

What each quantity means

$n(z)$ counts absorbing molecules per volume. For $1 \mu\text{g}/\text{m}^3$, $n(0) = (10^{-6} / 41.053) N_a = 1.46692 \times 10^{16} \text{ m}^{-3}$, with $N_a = 6.02214 \times 10^{23} \text{ mol}^{-1}$. The profile decreases with height.

$\sigma(f, z)$ is effective absorption area per molecule. Each line contributes $S \times V$: S is its integrated strength and V its shape across frequency. Temperature changes S and V . Pressure broadens and shifts V .

$$\alpha(f, z) = n(z)\sigma(f, z), \quad \tau(f) = \int_{z_g}^{z_s} \alpha(f, z) dz$$

α is absorption per unit distance. Optical depth τ adds it along the ray and has no units. Power transmission is $\exp(-\tau)$. At $\tau = 1$, 36.8% survives. Height enters through n , temperature, pressure and the endpoints.

Actual calculation at the first tone

$f_0 = 230.004883 \text{ GHz}$. At $z = 16.8826 \text{ m}$, $T = 288.0403 \text{ K}$ and $p = 101,122.35 \text{ Pa}$:

Calculation	Numerical value
One transition: $\sigma = S(T) \times V$	$(2.845759 \times 10^{-21}) \times 0.4334214 = 1.233413 \times 10^{-21} \text{ cm}^2$
All retained transitions	$\sigma_0 = 2.642246 \times 10^{-20} \text{ cm}^2/\text{molecule}$
Density for $1 \mu\text{g}/\text{m}^3$	$n_1(z) = 1.450501 \times 10^{10} \text{ molecules}/\text{cm}^3$
Integration weight	$w = 4,283.1123 \text{ cm}$
This node: $\delta\tau = \sigma_0 n_1 w$	$\delta\tau = 1.641537 \times 10^{-6}$
Sum all 114 altitude nodes	$\tau_{0,1} = 5.545084 \times 10^{-5} \text{ for } 1 \mu\text{g}/\text{m}^3$

97.17% of τ comes from below 5 km. Extra satellite height mainly adds free space loss.

The height fractions and cross sections depend on the assumed profile, frequency and atmosphere. They are calculated values.

3. The four numbers in the example

With the profile fixed, q scales the extra optical depth. The predicted channel change follows from that accumulated absorption.

Numerical chain at 230.004883 GHz

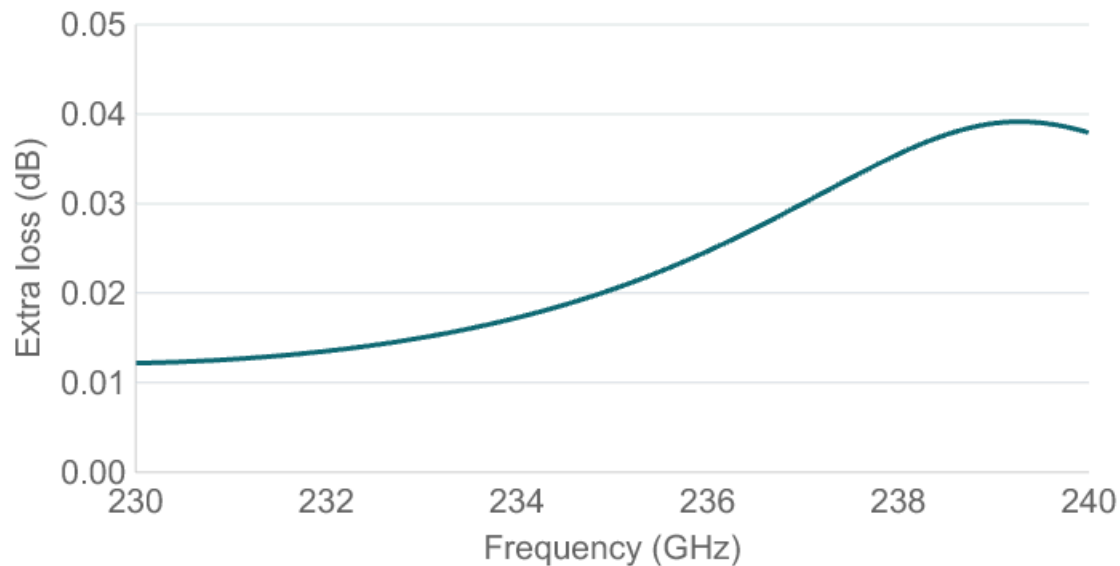
Quantity	Calculation and result
Chosen gas enhancement	$q = 50.5813966 \mu\text{g}/\text{m}^3$
Extra optical depth	$\tau_0 = q \times 5.5450845 \times 10^{-5} = 0.0028047812$
Extra molecular loss	$\Delta A_0 = (10 / \ln 10) \tau_0 = 0.0121810098 \text{ dB}$
Sample/reference amplitude	$\exp(-\tau_0/2) = 0.9985985923$
Amplitude reduction	$100 \times (1 - 0.9985985923) = 0.1401408\%$

$$\Delta A_k = a_k q, \quad a_0 = 0.0002408199585 \text{ dB}/(\mu\text{g}/\text{m}^3)$$

a_k means the extra loss produced by $1 \mu\text{g}/\text{m}^3$ with this profile. It varies with frequency because each tone samples a different part of the gas spectrum.

These are noiseless predictions for a chosen enhancement. The next page adds the modeled receiver errors.

The detector uses a pattern across frequency



The power transmission is $\exp(-\tau_0) = 0.99719915$: the added gas removes 0.2801% of the power. The amplitude change is smaller because power is proportional to magnitude squared.

Baseline oxygen/water loss at tone 0: 4.5477 dB.

4. Pilots: known symbols that reveal the channel

A pilot is a transmitted complex symbol whose value the receiver already knows. It provides a reference for estimating channel gain.

How one known symbol becomes a measurement

X is a known unit magnitude pilot, R the received symbol, h the channel and W thermal noise. k labels the tone and m the repetition. Divide out X before averaging:

$$R_{k,m} = h_k X_{k,m} + W_{k,m}$$
$$\hat{h}_k = (1/M) \sum_{m=1}^M R_{k,m} / X_{k,m}$$

Noiseless illustration with $h = 0.9985986$:

Known X	Received R	R / X
1	0.9985986	0.9985986
j	0.9985986 j	0.9985986

The second pilot rotates by 90°. Dividing by j removes that known rotation. Both pilots reveal the same channel, so their corrected estimates can be averaged.

Independent noise falls as $1/\sqrt{M}$ in standard deviation. Persistent gain drift survives averaging.

Conditional example: ideal tracking, matched background, 20 s total and an assumed, unmeasured 0.001 dB calibration residual.

The actual allocation and simulated observation

Resource	Value in this example
One OFDM symbol	102.4 ns useful + 10 ns CP = 112.4 ns
Pilot allocation	30 full pilot symbols per 10,000 symbols (0.3%)
Every tone, each 10 s	8,896 frames $\times$ 30 = M = 266,880 pilots
Thermal variance of $\hat{h}_0$	$1 / (M\rho_0) = 3.89892 \times 10^{-6}$ Pilot SNR $\rho_0 = 0.961036$

After tracking and normalization, tone 0 gives reference $\hat{h}_0$ and sample $\hat{h}_1$:

$$\hat{h}_{0,0} = 1.0009162403 + j 0.0004400251$$

$$\hat{h}_{1,0} = 1.0007326946 - j 0.0009412581$$

$$-20 \log_{10}(|\hat{h}_{1,0}|/|\hat{h}_{0,0}|) = 0.0015899416 \text{ dB}$$

5. Calibration error: a gain change can imitate gas

Calibration estimates and corrects the instrument response. The error is the response change left in the reference/sample comparison.

Two concrete examples

- 1. Receiver gain drops by 0.001 dB between acquisitions. With unchanged gas, the amplitude ratio becomes 0.9998848774 and the apparent loss is +0.001 dB.

At tone 0 alone, this would imply $0.001 / a_0 = 4.15248 \mu\text{g}/\text{m}^3$. If the error is constant across the band, our fit assigns it to gain offset b_0 instead of gas.
- 2. An error $e_k = a_k \times 1 \mu\text{g}/\text{m}^3$ has exactly the gas pattern. It imitates that concentration even across all tones. Independent calibration must constrain this ambiguity.

The residual in the saved simulation

Random thermal noise already affects the two pilot means. A separate persistent calibration draw contributes $e_0 = -0.0020537566$ dB:

$$y_0 = 0.0015899416 - 0.0020537566 = -0.0004638149 \text{ dB}$$

In measured data, calibration error is already present.

The 0.001 dB residual is an assumed standard deviation, not a measured accuracy or a maximum error. It does not shrink with pilot count.

The fit allows simple gain changes

$$y_k = a_k q + b_0 + b_1 u_k + \varepsilon_k, \quad A_{k,:} = [a_k \ 1 \ u_k]$$

Tone k	$a_k \times 10^4$	Offset	Slope u_k
0	2.408200	1	-0.500000
511	4.032712	1	-0.000489
1023	7.500802	1	0.500000

Only 3 example rows are shown; the fit uses all 1,024. a_k : dB/($\mu\text{g}/\text{m}^3$); u_k : -0.5 to +0.5. b_0 is a common shift, b_1 a tilt; ε is the remaining error.

$$C_{ij}(0) = \delta_{ij} v_i(0) + (0.001)^2 e^{-|f_i - f_j|/(10 \text{ GHz})}$$

$v_i(0) = (20 / \ln 10)^2 / (\text{Mpo}_i)$ includes both acquisitions. C is $1,024 \times 1,024$ in dB². δ_{ij} selects the diagonal. The second term models calibration errors shared across frequency.

6. Solve for concentration and make the decision

Fit $y_k = a_k q + b_0 + b_1 u_k + \epsilon_k$ to all 1,024 tones. q is the concentration increase ($\mu\text{g}/\text{m}^3$); b_0 and b_1 are gain offset and slope (dB).

$G = A^T C^{-1} A$ (3×3): weighted overlap between the gas, offset and slope templates.
 $g = A^T C^{-1} y$ (3×1): weighted match of those templates to the 1,024 observed losses.

Normal equation	Expanded equation: $G\theta = g$, with $\theta = [q, b_0, b_1]^T$
1. Concentration q	$0.301344020\ q + 458.463514\ b_0 + 127.082723\ b_1 = 15.066265191$
2. Gain offset b_0	$458.463513671\ q + 983654.579709\ b_0 - 223.779287\ b_1 = 22141.401031152$
3. Gain slope b_1	$127.082722926\ q - 223.779287\ b_0 + 198502.414349\ b_1 = 7023.360955313$

Minimize $J = (y - A\theta)^T C^{-1} (y - A\theta)$. Rows set $\partial J/\partial q = 0$, $\partial J/\partial b_0 = 0$ and $\partial J/\partial b_1 = 0$. Coefficients are rounded.

The estimate after fitting gain

Eliminating b_0 and b_1 leaves information $I_q = 0.006169586$ and weighted evidence $t_q = 0.243233919$ for the gas. Their ratio gives:

$$\hat{q} = t_q / I_q = 39.4247\ \mu\text{g}/\text{m}^3$$

The same solve gives $\hat{b}_0 = 0.00413651$ dB and $\hat{b}_1 = 0.01014643$ dB. The true input was $50.5814\ \mu\text{g}/\text{m}^3$. Observation errors explain the difference.

Conditional example: ideal tracking, matched background, 20 s total and an assumed, unmeasured 0.001 dB calibration residual.
No atmospheric detection measurement or 3D reconstruction is established by this example.

The threshold and its meaning

$$s_0 = I_q^{-1/2} = 12.73128\ \mu\text{g}/\text{m}^3$$

$$q_{\text{th}} = 2.32635\ s_0 = 29.61738\ \mu\text{g}/\text{m}^3$$

$\Phi(z) = P(Z \leq z)$, where Z is standard normal: mean 0, SD 1.
 $\Phi^{-1}(0.99) = 2.32635$: the cutoff with 99% below, 1% above.
Under $q = 0$, $\hat{q}/s_0$ is standard normal in the assumed model.

$39.4247 > 29.6174$: this draw detects an enhancement.
At $q = 50.5814$, predicted response is 95%, simulated 94.75%.
At $1\ \mu\text{g}/\text{m}^3$, simulated response is 1.31%.

7. The complete calculation in execution order

Inputs: atmospheric model, pilot observations and assumed error covariance. The arithmetic uses the full precision arrays.

1. Build the gas template from the layers

$n_1(z)$ is the molecular density for $q = 1 \mu\text{g}/\text{m}^3$. Sum absorption at every altitude node ℓ :

$$a_k = (10 / \ln 10) \sum_{\ell} \sigma(f_k, z_{\ell}) n_1(z_{\ell}) w_{\ell}$$

$$a_0 = 0.0002408199585 \text{ dB}/(\mu\text{g}/\text{m}^3).$$

2. Form the observed loss on each tone

$$\hat{h}_{r,k} = (1/M) \sum_m R_{r,k,m} / X_{k,m}, \quad r \in \{0, 1\}$$

$$y_k = -20 \log_{10}(|\hat{h}_{1,k}|/|\hat{h}_{0,k}|) + e_k$$

$r = 0$ reference, $r = 1$ sample. Add e only when simulating it. Here $y_0 = -0.0004638149 \text{ dB}$.

3. Compress 1,024 observations into G and g

$$A = [\mathbf{a} \ \mathbf{1} \ \mathbf{u}], \quad C = C(0) \text{ on page 5}$$

$$G = A^T C^{-1} A, \quad g = A^T C^{-1} \mathbf{y}$$

$\mathbf{u}_k = (f_k - \text{mean } f) / (\max f - \min f)$. G is 3×3 .

Conditional example: ideal tracking, matched background, 20 s total and an assumed, unmeasured 0.001 dB calibration residual.

The 1% threshold applies to this prespecified compound and the stated error model.

4. Eliminate the two gain parameters

Partition G and g into q and $b = [b_0, b_1]^T$:

$$I_q = G_{qq} - G_{qb} G_{bb}^{-1} G_{bq}$$

$$t_q = g_q - G_{qb} G_{bb}^{-1} g_b$$

$$I_q = 0.3013440205 - 0.2951744347 = 0.0061695858$$

$$t_q = 15.0662651908 - 14.8230312721 = 0.2432339187$$

5. Compute concentration and uncertainty

$$\hat{q} = t_q / I_q = 39.42467555 \mu\text{g}/\text{m}^3$$

$$s_0 = I_q^{-1/2} = 12.73127783 \mu\text{g}/\text{m}^3$$

s_0 is the standard deviation when $q = 0$.

6. Compare with the detection threshold

$$q_{\text{th}} = \Phi^{-1}(0.99) s_0 = 29.61738111 \mu\text{g}/\text{m}^3$$

$$\hat{q} = 39.4247 > q_{\text{th}} \Rightarrow \text{detection}$$

$\Phi^{-1}(0.99) = 2.32635$: 99% below, 1% above. See page 6.

8. Units for power, pilot noise and calibration

Absolute RF power, normalized complex pilot samples and attenuation errors describe different quantities and use different units.

Power and pilot noise at tone 0

Quantity	Value and unit
Total transmit power	25 dBm = 0.31623 W
Transmit power per tone	-5.1030 dBm
Received reference power	-97.5996 dBm
Noise in one tone	-97.4270 dBm
SNR $\rho_0 = P_{\text{signal}} / P_{\text{noise}}$	0.96104 linear = -0.17260 dB
Pilot-mean variance $E \delta h ^2 = 1/(M\rho_0)$	3.89892×10^{-6} Normalized, dimensionless
Thermal loss SD	0.01715 dB Reference + sample

Total covariance $C_{00} = 0.000295153 \text{ dB}^2$.
Pilot samples use linear complex amplitudes. Average R/X first, then take $-20 \log_{10}$ of the sample/reference magnitude ratio to obtain loss in dB.

The unit conversions are consistent. Achieving the assumed calibration stability remains unverified. Additional interference is not modeled here.

Why a calibration error can be tiny in dB

dBm specifies power relative to 1 mW. dB specifies a ratio. A drop from -97.5996 to -97.6006 dBm is 0.001 dB of extra loss.

Assumed residual SD	Relative power SD	Amplitude SD
0.001 dB (this example)	$\approx 0.0230\%$	$\approx 0.0115\%$
0.0001 dB (earlier design)	$\approx 0.00230\%$	$\approx 0.00115\%$

Small-error conversions: $\sigma P/P \approx (\ln 10 / 10) \sigma \text{dB}$ and $\sigma|h|/|h| \approx (\ln 10 / 20) \sigma \text{dB}$. These are standard deviations, not maximum errors or measured stability.

Interference needs a separate power model

For independent additive interference, add noise and interference powers in watts: $\text{SINR} = P_{\text{signal}} / (P_{\text{noise}} + P_{\text{interference}})$. Coherent interference can instead bias pilots. This example has no separate interference term.

9. Why use all 1,024 tones instead of one?

The extra loss changes with frequency. That pattern helps separate the gas contribution from a common gain shift and a gain slope.

One tone: one loss, three unknowns

At one frequency, the observation model is
 $y_k = a_k q + b_0 + b_1 u_k + \epsilon_k$.

A weaker signal can come from extra gas, a common gain change, or a frequency slope. One scalar observation cannot determine all three unknowns.

Even without noise, any chosen q and b_1 can be matched by setting $b_0 = y_k - a_k q - b_1 u_k$. Independent knowledge of the gain terms would be needed to infer q from that tone alone.

A plain average of all 1,024 losses gives one number. Gas and a common gain change can both shift that mean; averaging discards the spectral shape that helps distinguish them.

Conditional example: ideal tracking, matched background, 20 s total and an assumed, unmeasured 0.001 dB calibration residual. A calibration error with exactly the gas spectrum remains indistinguishable from gas, even with all tones.

Our special average: a weighted spectral fit

$\hat{q} = \sum_k w_k y_k$ uses all 1,024 tone losses.
Weights depend on the gas pattern a and error covariance C ; they need not sum to one.

Weight constraint	What it does
$\sum_k w_k a_k = 1$	Preserves the gas concentration q
$\sum_k w_k = 0$	Cancels a common gain offset b_0
$\sum_k w_k u_k = 0$	Cancels the gain slope b_1

Some weights are negative to cancel gain changes. Subject to these constraints, the fit chooses weights that minimize predicted variance $w^T C w$, accounting for noise and correlations.

Before this fit, each tone averages 266,880 pilots. More tones do not guarantee a $\sqrt{1,024}$ improvement: total 25 dBm power is split across the tones.